Kilian Koch

Doctoral Researcher in Mathematics
Chair of Applied Analysis · RWTH Aachen University

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I am a mathematician and doctoral researcher at the Chair of Applied Analysis at RWTH Aachen University. My research concerns gradient flows in geometric analysis, in particular well-posedness and singular behaviour of half-harmonic map heat flows, for instance due to topological constraints. My position is funded by the DFG through CRC 1481, project C01.

2 papers · 5 talks · 1 organized workshop · 8 teaching assignments

POSITIONS

Apr 2024 – Present
Doctoral Researcher
RWTH Aachen University
Chair of Applied Analysis. Funded by the German Research Foundation (DFG) via CRC 1481 *Sparsity and Singular Structures*, project C01.

Oct 2020 – Mar 2024
Mathematics Tutor
RWTH Aachen University
Held various tutoring positions, including Preliminary Course Tutor, Analysis 1-3 Tutor, Function Theory I Tutor, and Functional Analysis Tutor.

Feb 2018 – Present
Software Developer & Customer Service
Auto-Koch, Eschweiler
Developed the customer and vehicle management system KoKi and a scheduling program (desktop applications and locally used websites, JavaScript/Node.js). Regular customer service at the information desk and at exhibitions.

EDUCATION

Apr 2024 – Present (ongoing)
PhD in Mathematics
RWTH Aachen University

Oct 2022 – Apr 2024
M.Sc. Mathematics, with distinction (1.2)
RWTH Aachen University

Oct 2019 – Oct 2022
B.Sc. Mathematics, with distinction (1.3)
RWTH Aachen University

Aug 2016 – Aug 2019
High School Diploma (Abitur)
Mies van der Rohe School, Aachen

PUBLICATIONS & PREPRINTS

2026
Thresholding Scheme for the Half-Harmonic Map Heat Flow
Kilian Koch, Christof Melcher, Endre Süli
Preprint, arXiv:2609.25912

2025
Well-posedness of half-harmonic map heat flows for rough initial data
Kilian Koch, Christof Melcher
Preprint, arXiv:2504.06933

TALKS

4 December 2026
Well-posedness of half-harmonic map heat flows for rough initial data UPCOMING
NLPDE Seminar (invited talk), Kyoto University, Japan

30 September 2026
Thresholding Scheme for the Half-Harmonic Map Heat Flow
Annual Retreat of the CRC 1481 “Sparsity and Singular Structures”, Burg Frankenberg, Aachen
Slides

1 June 2026 — **Thresholding Scheme for the Half-Harmonic Map Heat Flow**
Young Researchers Workshop on Geometric Analysis, Singular PDEs and Numerics, RWTH Aachen University
Slides

4 December 2025 — **Well-posedness of Half-Harmonic Map Heat Flows for Rough Initial Data**
Applied Analysis Seminar (invited talk), Heidelberg University, Germany
Slides

1 October 2025 — **Well-posedness of half-harmonic map heat flows for rough initial data**
3 City Seminar Aachen–Dortmund–Eindhoven, TU Eindhoven, Netherlands
Slides

ORGANIZED EVENTS

1–3 June 2026 — **Young Researchers Workshop on Geometric Analysis, Singular PDEs and Numerics**
Organizer · Joint workshop of CRC 1481, EDDy and IntComSin (FAU Erlangen-Nürnberg / University of Regensburg), RWTH Aachen University

TEACHING EXPERIENCE

Aug 2026 – Sep 2026 — **Mathematics III | Co-lecturer**
International Academy RWTH Aachen, Aachen
Lecturer: Prof. Dr. rer. nat. Christof Melcher
Colleagues: Valentin Linse, M.Sc., Glal Bacho, M.Sc.
Tasks: Conducted interactive online exercise sessions for international students, provided academic support and created learning materials.
Partners: RWTH Aachen University, proRWTH-Freunde and Förderer der RWTH Aachen e.V.

Apr 2026 – Sep 2026 — **Höhere Mathematik 2 für Physiker und E-Techniker | Assistant**
RWTH Aachen, Aachen
Professor: Prof. Dr. Ivan Yaroslavtsev
Lecturer: Priv.-Doz. Dr. Olaf Wittich
Colleagues: Valentin Linse, M.Sc., Glal Bacho, M.Sc.
Tasks: Created exercises, held exercise sessions, corrected exams, and prepared exam questions.

Oct 2025 – Mar 2026 — **Höhere Mathematik 1 für Physiker und E-Techniker | Assistant**
RWTH Aachen, Aachen
Professor: Prof. Dr. Ivan Yaroslavtsev
Lecturer: Priv.-Doz. Dr. Olaf Wittich
Colleagues: Valentin Linse, M.Sc., Glal Bacho, M.Sc.
Tasks: Created exercises, held exercise sessions, corrected exams, and prepared exam questions.

Aug 2025 – Sep 2025 — **Mathematics III | Co-lecturer**
International Academy RWTH Aachen, Aachen
Lecturer: Prof. Dr. rer. nat. Christof Melcher
Colleagues: Valentin Linse, M.Sc., Glal Bacho, M.Sc.
Tasks: Conducted interactive online exercise sessions for international students, provided academic support and created learning materials.
Partners: RWTH Aachen University, proRWTH-Freunde and Förderer der RWTH Aachen e.V.

Apr 2025 – Sep 2025 — **Partielle Differentialgleichungen 1 | Assistant**
RWTH Aachen, Aachen
Professor: Prof. Dr. rer. nat. Christof Melcher
Colleague: Glal Bacho, M.Sc.
Tasks: Created exercises, held exercise sessions, corrected exams, and prepared exam questions.

Oct 2024 – Mar 2025 — **Analysis 3 für Mathematiker | Assistant**
RWTH Aachen, Aachen
Professor: Prof. Dr. rer. nat. Christof Melcher
Colleague: Glal Bacho, M.Sc.
Tasks: Created exercises, held exercise sessions, corrected exams, and prepared exam questions.

Aug 2024 – Sep 2024 — **Mathematics III | Co-lecturer**
International Academy RWTH Aachen, Aachen

Lecturer: Prof. Dr. rer. nat. Christof Melcher
Colleagues: Georg Heinen, M.Sc., Glal Bacho, M.Sc.
Tasks: Conducted interactive online exercise sessions for international students, provided academic support and created learning materials.
Partners: RWTH Aachen University, proRWTH-Freunde and Förderer der RWTH Aachen e.V.

Apr 2024 – Sep 2024 **Höhere Mathematik 2 für Physiker und E-Techniker | Assistant**
RWTH Aachen, Aachen
Professor: Prof. Dr. Sebastian Noelle
Lecturer: Priv.-Doz. Dr. Olaf Wittich
Colleague: Tom Goertzen, Postdoctoral Research Associate
Tasks: Developed digital exercises, conducted exercise sessions, and graded exams.

RESEARCH FUNDING

- Doctoral researcher position funded by the German Research Foundation (DFG) through CRC 1481 *Sparsity and Singular Structures* (project no. 442047500), project C01 *Singularity formation in dissipative harmonic flows*, since October 2024

AWARDS

- RWTH Dean's List, Winter Semester 2021/22 - Summer Semester 2022
- RWTH Dean's List, Winter Semester 2020/21 - Summer Semester 2021
- RWTH Dean's List, Winter Semester 2019/20 - Summer Semester 2020

SKILLS

- **Programming:** JavaScript/Node.js, C/C++, Python, HTML/CSS; basics of Haskell, Prolog and Java
- **Languages:** German (native), English (C1), Japanese (JLPT N3, ≈ B1)